

CHAPTER 1

Motivation: Challenges and Opportunities for BDI

The big data era is the inevitable consequence of *datafication*: our ability to transform each event and every interaction in the world into digital data, and our concomitant desire to analyze and extract value from this data. Big data comes with a lot of promise, enabling us to make valuable, data-driven decisions to alter all aspects of society.

Big data is being generated and used today in a variety of domains, including data-driven science, telecommunications, social media, large-scale e-commerce, medical records and e-health, and so on. Since the value of data explodes when it can be linked and fused with other data, addressing the *big data integration* (BDI) challenge is critical to realizing the promise of big data in these and other domains.

As one prominent example, recent efforts in mining the web and extracting entities, relationships, and ontologies to build general purpose knowledge bases such as Freebase [Bollacker et al. 2008], the Google knowledge graph [Dong et al. 2014a], ProBase [Wu et al. 2012], and Yago [Weikum and Theobald 2010] show promise of using integrated big data to improve applications such as web search and web-scale data analysis.

As a second important example, the flood of geo-referenced data available in recent years, such as geo-tagged web objects (e.g., photos, videos, tweets), online check-ins (e.g., Foursquare), WiFi logs, GPS traces of vehicles (e.g., taxi cabs), and roadside sensor networks has given momentum for using such integrated big data to characterize large-scale human mobility [Becker et al. 2013], and influence areas like public health, traffic engineering, and urban planning.

In this chapter, we first describe the problem of data integration and the components of traditional data integration in Section 1.1. We then discuss the specific challenges that arise in BDI in Section 1.2, where we first identify the dimensions along which BDI differs from traditional data integration, then present a number of recent case studies that empirically study the nature of data sources in BDI. BDI also offers opportunities that do not exist in traditional data integration, and we highlight some of these opportunities in Section 1.3. Finally, we present an outline of the rest of the book in Section 1.4.

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1.1 TRADITIONAL DATA INTEGRATION

Data integration has the goal of providing *unified access to data* residing in multiple, autonomous data sources. While this goal is easy to state, achieving this goal has proven notoriously hard, even for a small number of sources that provide structured data—the scenario of traditional data integration [Doan et al. 2012].

To understand some of the challenging issues in data integration, consider an illustrative example from the Flights domain, for the common tasks of tracking flight departures and arrivals, examining flight schedules, and booking flights.

1.1.1 THE FLIGHTS EXAMPLE: DATA SOURCES

We have a few different kinds of sources, including two airline sources *Airline1* and *Airline2* (e.g., United Airlines, American Airlines, Delta, etc.), each providing flight data about a different airline, an airport source *Airport3*, providing information about flights departing from and arriving at a particular airport (e.g., EWR, SFO), a comparison shopping travel source *Airfare4* (e.g., Kayak, Orbitz, etc.), providing fares in different fare classes to compare alternate flights, and an informational source *Airinfo5* (e.g., a Wikipedia table), providing data about airports and airlines.

Sample data for the various source tables is shown in Tables 1.1–1.8, using short attribute names for brevity. The mapping between the short and full attribute names is provided in Table 1.9 for ease of understanding. Records in different tables that are highlighted using the same color are related to each other, and the various tables should be understood as follows.

Source Airline1

Source *Airline1* provides the tables *Airline1.Schedule*(Flight Id, Flight Number, Start Date, End Date, Departure Time, Departure Airport, Arrival Time, Arrival Airport) and *Airline1.Flight*(Flight Id, Departure Date, Departure Time, Departure Gate, Arrival Date, Arrival Time, Arrival Gate, Plane Id). The underlined attributes form a key for the corresponding table, and *Flight Id* is used as a join key between these two tables.

Table *Airline1.Schedule* shows flight schedules in Table 1.1. For example, record r_{11} in table *Airline1.Schedule* states that *Airline1*’s flight 49 is scheduled to fly regularly from *EWR* to *SFO*, departing at 18:05, and arriving at 21:10, between 2013-10-01 and 2014-03-31. Record r_{12} in the same table shows that the same flight 49 has different scheduled departure and arrival times between 2014-04-01 and 2014-09-30. Records r_{13} and r_{14} in the same table show the schedules for two different segments of the same flight 55, the first from *ORD* to *BOS*, and the second from *BOS* to *EWR*, between 2013-10-01 and 2014-09-30.

Table *Airline1.Flight* shows the actual departure and arrival information in Table 1.2, for the flights whose schedules are shown in *Airline1.Schedule*. For example, record r_{21} in table *Airline1.Flight*

TABLE 1.1: Sample data for Airline1.Schedule

	FI	FN	SD	ED	DT	DA	AT	AA
r_{11}	123	49	2013-10-01	2014-03-31	18:05	EWR	21:10	SFO
r_{12}	234	49	2014-04-01	2014-09-30	18:20	EWR	21:25	SFO
r_{13}	345	55	2013-10-01	2014-09-30	18:30	ORD	21:30	BOS
r_{14}	346	55	2013-10-01	2014-09-30	22:30	BOS	23:30	EWR

TABLE 1.2: Sample data for Airline1.Flight

	FI	DD	DT	DG	AD	AT	AG	PI
r_{21}	123	2013-12-21	18:45	C98	2013-12-21	21:30	81	4013
r_{22}	123	2013-12-28	21:30	C101	2013-12-29	00:30	81	3008
r_{23}	345	2013-12-29	18:30	B6	2013-12-29	21:45	C18	4013
r_{24}	346	2013-12-29	22:35	C18	2013-12-29	23:35	C101	4013

records information about a specific flight, corresponding to the regularly scheduled flight r_{11} (the Flight Id 123 specifies the join key), using a plane with id 4013, actually departing on 2013-12-21 at 18:45 (40 minutes later than the scheduled departure time of 18:05) from gate C98, and actually arriving on 2013-12-21 at 21:30 (20 minutes later than the scheduled arrival time of 21:10) at gate 81. Both r_{11} and r_{21} use yellow highlighting to visually depict their relationship. Record r_{22} in the same table records information about a flight on a different date, also corresponding to the regularly scheduled flight r_{11} , with a considerably longer delay in departure and arrival times. Records r_{23} and r_{24} record information about flights on 2013-12-29, corresponding to regularly scheduled flights r_{13} and r_{14} , respectively.

Source Airline2

Source Airline2 provides similar data to source Airline1, but using the table Airline2.Flight(Flight Number, Departure Airport, Scheduled Departure Date, Scheduled Departure Time, Actual Departure Time, Arrival Airport, Scheduled Arrival Date, Scheduled Arrival Time, Actual Arrival Time).

Each record in table Airline2.Flight, shown in Table 1.3, contains both the schedule and the actual flight details. For example, record r_{31} records information about Airline2's flight 53, departing from SFO, scheduled to depart on 2013-12-21 at 15:30, with a 30 minute delay in the actual departure time, arriving at EWR, scheduled to arrive on 2013-12-21 at 23:35, with a 40 minute

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TABLE 1.3: Sample data for Airline2.Flight

	FN	DA	SDD	SDT	ADT	AA	SAD	SAT	AAT
r_{31}	53	SFO	2013-12-21	15:30	16:00	EWB	2013-12-21	23:35	00:15 (+1d)
r_{32}	53	SFO	2013-12-22	15:30	16:15	EWB	2013-12-22	23:35	00:30
r_{33}	53	SFO	2014-06-28	16:00	16:05	EWB	2014-06-29	00:05	23:57 (-1d)
r_{34}	53	SFO	2014-07-06	16:00	16:00	EWB	2014-07-07	00:05	00:09
r_{35}	49	SFO	2013-12-21	12:00	12:35	EWB	2013-12-21	20:05	20:45
r_{36}	77	LAX	2013-12-22	09:15	09:15	SFO	2013-12-22	11:00	10:59

TABLE 1.4: Sample data for Airport3.Departures

	AL	FN	S	A	GT	TT	T	G	R
r_{41}	A1	49	2013-12-21	2013-12-21	18:45	18:53	C	98	2
r_{42}	A1	49	2013-12-28	2013-12-28	21:29	21:38	C	101	2

delay in the actual arrival time; its arrival on *2013-12-22* (the day after its scheduled arrival) is indicated by the *(+1d)* associated with the actual arrival time. Note that this table contains a record r_{35} for Airline2's flight *49*, which is different from Airline1's flight *49*, illustrating that different airlines can use the same flight number for their respective flights.

Unlike source Airline1, source Airline2 does not publish the departure gate, arrival gate, and the plane identifier used for the specific flight, illustrating the diversity between the schemas used by these sources.

Source Airport3

Source Airport3 provides tables Airport3.Departures(Air Line, Flight Number, Scheduled, Actual, Gate Time, Takeoff Time, Terminal, Gate, Runway) and Airport3.Arrivals(Air Line, Flight Number, Scheduled, Actual, Gate Time, Landing Time, Terminal, Gate, Runway).

Table Airport3.Departures, shown in Table 1.4, publishes information only about flight departures from *EWB*. For example, record r_{41} in table Airport3.Departures states that Airline1's flight *49*, scheduled to depart on *2013-12-21*, departed on *2013-12-21* from terminal *C* and gate *98* at *18:45* and took off at *18:53* from runway *2*. There is no information in this table about the arrival airport, arrival date, and arrival time of this flight. Note that r_{41} corresponds to records r_{11} and r_{21} , depicted by the consistent use of the yellow highlight.

TABLE 1.5: Sample data for Airport3.Arrivals

	AL	FN	S	A	GT	LT	T	G	R
r_{51}	A2	53	2013-12-21	2013-12-22	00:21	00:15	B	53	2
r_{52}	A2	53	2013-12-22	2013-12-23	00:40	00:30	B	53	2
r_{53}	A1	55	2013-12-29	2013-12-29	23:35	23:31	C	101	1
r_{54}	A2	49	2013-12-21	2013-12-21	20:50	20:45	B	55	2

Table Airport3.Arrivals, shown in Table 1.5, publishes information only about flight arrivals into *EWB*. For example, record r_{51} in table Airport3.Arrivals states that Airline2's flight 53, scheduled to arrive on 2013-12-21, arrived on 2013-12-22, landing on runway 2 at 00:15, reaching gate 53 of terminal B at 00:21. There is no information in this table about the departure airport, departure date, and departure time of this flight. Note that r_{51} corresponds to record r_{31} , both of which are highlighted in lavender.

Unlike sources Airline1 and Airline2, source Airport3 distinguishes between the time at which the flight left/reached the gate and the time at which the flight took off from/landed at the airport runway.

Source Airfare4

Travel source Airfare4 publishes comparison shopping data for multiple airlines, including schedules in Airfare4.Flight(Flight Id, Flight Number, Departure Airport, Departure Date, Departure Time, Arrival Airport, Arrival Time) and fares in Airfare4.Fares(Flight Id, Fare Class, Fare). Flight Id is used as a join key between these two tables.

For example, record r_{61} in Airfare4.Flight, shown in Table 1.6, states that Airline1's flight A1-49 was scheduled to depart from Newark Liberty airport on 2013-12-21 at 18:05, and arrive at the San Francisco airport on the same date at 21:10. Note that r_{61} corresponds to records r_{11} , r_{21} , and r_{41} , indicated by the yellow highlight shared by all records.

The records in table Airfare4.Fares, shown in Table 1.7, gives the fares for various fare classes of this flight. For example, record r_{71} shows that fare class A of this flight has a fare of \$5799.00; the flight identifier 456 is the join key.

Source Airinfo5

Informational source Airinfo5 publishes data about airports and airline in Airinfo5.AirportCodes(Airport Code, Airport Name) and Airinfo5.AirlineCodes(Air Line Code, Air Line Name), respectively.

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TABLE 1.6: Sample data for Airfare4.Flight

	FI	FN	DA	DD	DT	AA	AT
r_{61}	456	A1-49	Newark Liberty	2013-12-21	18:05	San Francisco	21:10
r_{62}	457	A1-49	Newark Liberty	2014-04-05	18:05	San Francisco	21:10
r_{63}	458	A1-49	Newark Liberty	2014-04-12	18:05	San Francisco	21:10
r_{64}	460	A2-53	San Francisco	2013-12-22	15:30	Newark Liberty	23:35
r_{65}	461	A2-53	San Francisco	2014-06-28	15:30	Newark Liberty	23:35
r_{66}	462	A2-53	San Francisco	2014-07-06	16:00	Newark Liberty	00:05 (+1d)

TABLE 1.7: Sample data for Airfare4.Fares

	FI	FC	F
r_{71}	456	A	\$5799.00
r_{72}	456	K	\$999.00
r_{73}	456	Y	\$599.00

TABLE 1.8: Sample data for Airinfo5.AirportCodes, Airinfo5.AirlineCodes

	Airinfo5.AirportCodes			Airinfo5.AirlineCodes	
	AC	AN		ALC	ALN
r_{81}	EWB	Newark Liberty, NJ, US	r_{91}	A1	Airline1
r_{82}	SFO	San Francisco, CA, US	r_{92}	A2	Airline2

For example, record r_{81} in Airinfo5.AirportCodes, shown in Table 1.8, states that the name of the airport with code *EWB* is *Newark Liberty, NJ, US*. Similarly, record r_{91} in Airinfo5.AirlineCodes, also shown in Table 1.8, states that the name of the airline with code *A1* is *Airline1*.

1.1.2 THE FLIGHTS EXAMPLE: DATA INTEGRATION

While each of the five sources is useful in isolation, the value of this data is considerably enhanced when the different sources are integrated.

TABLE 1.9: Abbreviated attribute names

Short Name	Full Name	Short Name	Full Name
A	Actual	AA	Arrival Airport
AAT	Actual Arrival Time	AC	Airport Code
AD	Arrival Date	ADT	Actual Departure Time
AG	Arrival Gate	AL	Air Line
ALC	Air Line Code	ALN	Air Line Name
AN	Airport Name	AT	Arrival Time
DA	Departure Airport	DD	Departure Date
DG	Departure Gate	DT	Departure Time
ED	End Date	F	Fare
FC	Fare Class	FI	Flight Id
FN	Flight Number	G	Gate
GT	Gate Time	LT	Landing Time
PI	Plane Id	R	Runway
S	Scheduled	SAD	Scheduled Arrival Date
SAT	Scheduled Arrival Time	SD	Start Date
SDD	Scheduled Departure Date	SDT	Scheduled Departure Time
T	Terminal	TT	Takeoff Time

Integrating Sources

First, each airline source (e.g., *Airline1*, *Airline2*) benefits by linking with the airport source *Airport3* since the airport source provides much more detailed information about the actual flight departures and arrivals, such as gate time, takeoff and landing times, and runways used; this can help the airlines better understand the reasons for flight delays. Second, airport source *Airport3* benefits by linking with the airline sources (e.g., *Airline1*, *Airline2*) since the airline sources provide more detailed information about the flight schedules and overall flight plans (especially for multi-hop flights such as *Airline1*'s flight 55); this can help the airport better understand flight patterns. Third, the comparison shopping travel source *Airfare4* benefits by linking with the airline and airport sources to provide additional information such as historical on-time departure/arrival statistics; this can be very useful to customers as they make flight bookings. This linkage makes critical use of the informational source *Airinfo5*, as we shall see later. Finally, customers benefit when the various sources are integrated since they do not need to go to multiple sources to obtain all the information they need.

For example, the query “*for each airline flight number, compute the average delays between scheduled and actual departure times, and between actual gate departure and takeoff times, over the past one month*” can be easily answered over the integrated database, but not using any single source.

However, integrating multiple, autonomous data sources can be quite difficult, often requiring considerable manual effort to understand the semantics of the data in each source to resolve ambiguities. Consider, again, our illustrative Flights example.

Semantic Ambiguity

In order to *align* the various source tables correctly, one needs to understand that (i) the same conceptual information may be modeled quite differently in different sources, and (ii) different conceptual information may be modeled similarly in different sources.

For example, source Airline1 models schedules in table Airline1.Schedule within date ranges (specified by Start Date and End Date), using attributes Departure Time and Arrival Time for time information. However, source Airline2 models schedules along with actual flight information in the table Airline2.Flight, using different records for different actual flights, and differently named attributes Scheduled Departure Date, Scheduled Departure Time, Scheduled Arrival Date, and Scheduled Arrival Time.

As another example, source Airport3 models both actual gate departure/arrival times (Gate Time in Airport3.Departures and Airport3.Arrivals) and actual takeoff/landing times (Takeoff Time in Airport3.Departures, Landing Time in Airport3.Arrivals). However, each of Airline1 and Airline2 models only one kind of departure and arrival times; in particular, a careful examination of the data shows that source Airline1 models gate times (Departure Time and Arrival Time in Airline1.Schedule and Airline1.Flight) and Airline2 models takeoff and landing times (Scheduled Departure Time, Actual Departure Time, Scheduled Arrival Time, Actual Arrival Time in Airline2.Flight).

To illustrate that different conceptual information may be modeled similarly, note that Departure Date is used by source Airline1 to model actual departure date (in Airline1.Flight), but is used to model scheduled departure date by source Airfare4 (in Airfare4.Flight).

Instance Representation Ambiguity

In order to *link* the same data instance from multiple sources, one needs to take into account that instances may be represented differently, reflecting the autonomous nature of the sources.

For example, flight numbers are represented in sources Airline1 and Airline2 using digits (e.g., 49 in r_{11} , 53 in r_{31}), while they are represented in source Airfare4 using alphanumerics (e.g., A1-49 in r_{61}). Similarly, the departure and arrival airports are represented in sources Airline1 and Airline2 using 3-letter codes (e.g., EWR, SFO, LAX), but as a descriptive string in Airfare4.Flight (e.g., Newark Liberty, San Francisco). Since flights are uniquely identified by the combination of attributes (Airline, Flight Number, Departure Airport, Departure Date), one would not be able to link the data in Airfare4.Flight

with the corresponding data in *Airline1*, *Airline2*, and *Airport3* without additional tables mapping airline codes to airline descriptive names, and airport codes to airport descriptive names, such as *Airinfo5.AirlineCodes* and *Airinfo5.AirportCodes* in Table 1.8. Even with such tables, one might need approximate string matching techniques [Hadjieleftheriou and Srivastava 2011] to match *Newark Liberty* in *Airfare4.Flight* with *Newark Liberty, NJ, US* in *Airinfo5.AirportCodes*.

Data Inconsistency

In order to *fuse* the data from multiple sources, one needs to resolve the instance-level ambiguities and inconsistencies between the sources.

For example, there is an inconsistency between records r_{32} in *Airline2.Flight* and r_{52} in *Airport3.Arrivals* (both of which are highlighted in blue to indicate that they refer to the same flight). Record r_{32} states that the Scheduled Arrival Date and Actual Arrival Time of *Airline2*'s flight 53 are 2013-12-22 and 00:30, respectively, implying that the actual arrival date is the same as the scheduled arrival date (unlike record r_{31} , where the Actual Arrival Time included *(+1d)* to indicate that the actual arrival date was the day after the scheduled arrival date). However, r_{52} states this flight arrived on 2013-12-23 at 00:30. This inconsistency would need to be resolved in the integrated data.

As another example, record r_{62} in *Airfare4.Flight* states that *Airline1*'s flight 49 on 2014-04-05 is scheduled to depart and arrive at 18:05 and 21:10, respectively. While the departure date is consistent with record r_{12} in *Airline1.Schedule* (both r_{12} and r_{62} are highlighted in green to indicate their relationship), the scheduled departure and arrival times are not, possibly because r_{62} incorrectly used the (out-of-date) times from r_{11} in *Airline1.Schedule*. Similarly, record r_{65} in *Airfare4.Flight* states that *Airline2*'s flight 53 on 2014-06-28 is scheduled to depart and arrive at 15:30 and 23:35, respectively. While the departure date is consistent with record r_{33} in *Airline2.Flight* (both r_{33} and r_{65} are highlighted in greenish yellow to indicate their relationship), the scheduled departure and arrival times are not, possibly because r_{65} incorrectly used the out-of-date times from r_{32} in *Airline2.Flight*. Again, these inconsistencies need to be resolved in the integrated data.

1.1.3 DATA INTEGRATION: ARCHITECTURE & THREE MAJOR STEPS

Traditional data integration addresses these challenges of semantic ambiguity, instance representation ambiguity, and data inconsistency by using a pipelined architecture, which consists of three major steps, depicted in Figure 1.1.

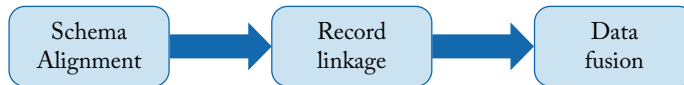


FIGURE 1.1: Traditional data integration: architecture.

The first major step in traditional data integration is that of *schema alignment*, which addresses the challenge of semantic ambiguity and aims to understand which attributes have the same meaning and which ones do not. More formally, we have the following definition.

Definition 1.1 (Schema Alignment) Consider a set of source schemas in the same domain, where different schemas may describe the domain in different ways. *Schema alignment* generates three outcomes.

1. A *mediated schema* that provides a unified view of the disparate sources and captures the salient aspects of the domain being considered.
2. An *attribute matching* that matches attributes in each source schema to the corresponding attributes in the mediated schema.
3. A *schema mapping* between each source schema and the mediated schema to specify the semantic relationships between the contents of the source and that of the mediated data.

The result schema mappings are used to reformulate a user query into a set of queries on the underlying data sources for query answering.

This step is non-trivial for many reasons. Different sources can describe the same domain using very different schemas, as illustrated in our Flights example. They may use different attribute names even when they have the same meaning (e.g., *Arrival Date* in *Airline1.Flight*, *Actual Arrival Date* in *Airline2.Flight*, and *Actual* in *Airport3.Arrivals*). Also, sources may apply different meanings for attributes with the same name (e.g., *Actual* in *Airport3.Departures* refers to the actual departure date, while *Actual* in *Airport3.Arrivals* refers to the actual arrival date).

The second major step in traditional data integration is that of *record linkage*, which addresses the challenge of instance representation ambiguity, and aims to understand which records represent the same entity and which ones do not. More formally, we have the following definition.

Definition 1.2 (Record Linkage) Consider a set of data sources, each providing a set of records over a set of attributes. *Record linkage* computes a partitioning of the set of records, such that each partition identifies the records that refer to a distinct entity.

Even when schema alignment has been performed, this step is still challenging for many reasons. Different sources can describe the same entity in different ways. For example, records r_{11} in *Airline1.Schedule* and r_{21} in *Airline1.Flight* should be linked to record r_{41} in *Airport3.Departures*; however, r_{11} and r_{21} do not explicitly mention the name of the airline, while r_{41} does not explicitly mention the departure airport, both of which are needed to uniquely identify a flight. Further, different sources may use different ways of representing the same information (e.g., the alternate ways of representing airports as discussed earlier). Finally, comparing every pair of records to determine whether or not they refer to the same entity can be infeasible in the presence of billions of records.

The third major step in traditional data integration is that of *data fusion*, which addresses the challenge of data quality, and aims to understand which value to use in the integrated data when the sources provide conflicting values. More formally, we have the following definition.

Definition 1.3 (Data Fusion) Consider a set of data items, and a set of data sources each of which provides values for a subset of the data items. *Data fusion* decides the true value(s) for each data item.

Such conflicts can arise for a variety of reasons including mis-typing, incorrect calculations (e.g., the conflict in actual arrival dates between records r_{32} and r_{52}), out-of-date information (e.g., the conflict in scheduled departure and arrival times between records r_{12} and r_{62}), and so on.

We will describe approaches used for each of these steps in subsequent chapters, and move on to highlighting the challenges and opportunities that arise when moving from traditional data integration to big data integration.

1.2 BDI: CHALLENGES

To appreciate the challenges that arise in big data integration, we present five recent case studies that empirically examined various characteristics of data sources on the web that would be integrated in BDI efforts, and the dimensions along which these characteristics are naturally classified.

When you can measure what you are speaking about, and express it in numbers, you know something about it. —Lord Kelvin

1.2.1 THE “V” DIMENSIONS

Big data integration differs from traditional data integration along many dimensions, paralleling the dimensions along which big data is characterized as differing from traditional databases.

Volume

In the big data era, not only can data sources contain a huge volume of data, but also the number of data sources has grown to be in the millions; even for a single domain, the number of sources has grown to be in the tens to hundreds of thousands.

There are many scenarios where a single data source can contain a huge volume of data, ranging from social media and telecommunications networks to finance.

To illustrate a scenario with a large number of sources in a single domain, consider again our Flights example. Suppose we would like to extend it to all airlines and all airports in the world to support flexible, international travel itineraries. With hundreds of airlines worldwide, and over